

Date: **22 MAY 2025**

Regd. No.

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Sub Code: **23EE/EC/CS2T02****MIC23**

B. Tech II Semester Regular/Supplementary Examinations, May-2025

CHEMISTRY

(Common to EEE, ECE, CSE)

Time: 3 hours**Max Marks: 70M**

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART – A			10 X 2 = 20 M		
			Marks	BTL	CO
I.	a)	Draw the π –molecular orbitals of butadiene.	2 M	L3	CO1
	b)	On the basis of molecular orbital theory, explain that O ₂ is paramagnetic.	2 M	L2	CO1
	c)	How supercapacitor is advantageous over a battery?	2 M	L2	CO2
	d)	Write any four differences between type-1 and type-2 superconductors.	2 M	L3	CO2
	e)	Distinguish between primary and secondary batteries.	2 M	L4	CO3
	f)	What are amperometric sensors and write the examples of it.	2 M	L2	CO3
	g)	Write the preparation of the Nylon-6,6.	2 M	L2	CO4
	h)	Mention the applications of PGA in various fields.	2 M	L3	CO4
	i)	Write about the types of modes in IR.	2 M	L2	CO5
	j)	Write any two electronic transitions in UV spectroscopy.	2 M	L3	CO5
PART – B			5 X 10 = 50 M		
UNIT – I					
1.	a)	Derive Schrodinger wave equation of a particle in three dimensional space.	5M	L4	CO1
	b)	Outline the Postulates of Molecular orbital theory by using molecular orbital diagram with a suitable example.	5M	L2	CO1
(OR)					
2.	a)	Construct π –molecular orbitals of Benzene and arrange the orbitals as per their energy levels and explain their formation.	5M	L4	CO1
	b)	Make use of the concept of the molecular orbital theory to draw the molecular energy level diagram of CO, Calculate B.O and mention its magnetic properties.	5M	L3	CO1
UNIT – II					
3.	a)	Summarize the applications of superconductors and mention the examples of superconductors with their critical temperatures.	5M	L2	CO2
	b)	Classify the types of fullerenes , write down their properties and applications.	5M	L4	CO2
(OR)					
4.	a)	What is Hierarchical Porous Carbon (HPC)? List its applications.	5M	L2	CO2
	b)	What is a supercapacitor? Explain about the Hybrid capacitor.	5M	L2	CO2
UNIT – III					
5.	a)	Discuss the conductometric titrations in the following cases. a. Strong acid and Weak base b. Weak acid and strong base	5M	L2	CO3
	b)	What is a fuel cell? Explain the construction, working and applications of polymer electrolyte membrane fuel cell.	5M	L2	CO3

(OR)					
6.	a)	What will be standard cell potential (E^0 cell) of galvanic cell with the following reaction? $2\text{Cr(s)} + 3\text{Cd}^{2+}(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{Cd(s)}$ $[\text{E}^0_{\text{Cr}^{3+}/\text{Cr}} = -0.74\text{V and } \text{E}^0_{\text{Cd}^{2+}/\text{Cd}} = -0.40\text{V}]$	5M	L2	CO3
	b)	Outline the construction and working of Zinc-air battery. List its uses.	5M	L4	CO3
UNIT – IV					
7.	a)	What is a biodegradable polymer? Write short notes on the preparation, properties, and applications of polyglycolic acid.	5M	L2	CO4
	b)	Outline the preparation, properties, and applications of Bakelite.	5M	L2	CO4
(OR)					
8.	a)	Distinguish thermoplastic and Thermosetting plastic.	5M	L4	CO4
	b)	Explain the preparation, properties, and applications of Teflon.	5M	L2	CO4
UNIT – V					
9.	a)	Explain about classification and applications of chromatography.	5M	L2	CO5
	b)	List out the applications of UV Spectroscopy.	5M	L4	CO5
(OR)					
10.	a)	Discuss the principle and instrumentation of UV spectroscopy.	5M	L4	CO5
	b)	What is the principle involved in IR spectroscopy? List its applications.	5M	L2	CO5

***** End of the Question Paper *****